1] Association learning:

It is an unsupervised ML ~~alg~~ technique used to discover interesting relationships (pattern) b/w variables in large datasets.

→ commonly used in market basket analysis

→ Finds if-then relationship

Ex: If a person buys bread → they ~~also~~ buy butter

Types: (1) Apriori
(2) Eclat
(3) F-P growth

2] Apriori Algorithm: - Uses frequent itemsets
- Based on Apriori property:
If an itemset is freq, all its subsets must also be frequent

Step 1: Generate candidate item sets

Step 2: Calculate support
Keep only itemsets with support ≥ minimum threshold
$L_k = \{ C_k \mid support \geq min_support \}$

Step 3: Generate Next candidates, $C_{k+1} = L_k \times L_k$

Step 4: Repeat until no more frequent itemsets

Key Terms :-

Antecedent : The 'if' part of the rule

Consequent : The 'then' part of the rule

Rule :- Bread → Butter

3] ECLAT Algorithm :

Equivalence Class Clustering and bottom-up Lattice Traversal

- Algorithm that uses a vertical data format

Key Idea : - Uses Transaction ID sets (TID sets)

- Finds frequent item sets using set intersection

Example : Milk → {1, 2, 4}

Bread → {1, 3, 4}

Milk ∩ Bread = {1, 4}

Adv : Faster than Apriori for large datasets

4] FP - Growth Algorithm :

FP (Frequent Pattern) - Growth that uses a tree structure (FP - Tree) to find frequent patterns without generating candidate sets.

How it works :

(1) Scan dataset → find frequent items

(2) Build FP - Tree

(3) Extract frequent patterns recursively

Example use case

Supermarket analysis :

- If customers buy Milk + Bread, suggests Butter
- Used in Amazon recommendations

5) Evaluation metrics:

(1) Support : $\text{Support}(A \to B) = \dfrac{\text{Transactions containing } A \,\&\, B}{\text{Total transactions}}$

(2) confidence : $\text{confidence}(A \to B) = \dfrac{\text{Support}(A \cap B)}{\text{Support}(A)}$

(3) Lift : $\text{Lift}(A \to B) = \dfrac{\text{Confidence}(A \to B)}{\text{Support}(B)}$

Lift > 1 → strong rule

6) What is Feature engineering ?

It is the process of creating, transforming & selecting features from raw data to improve model performance.